Charting New Paths in Lipidomics: An Evaluation of DDA and DIA on an Astral Mass Spectrometer

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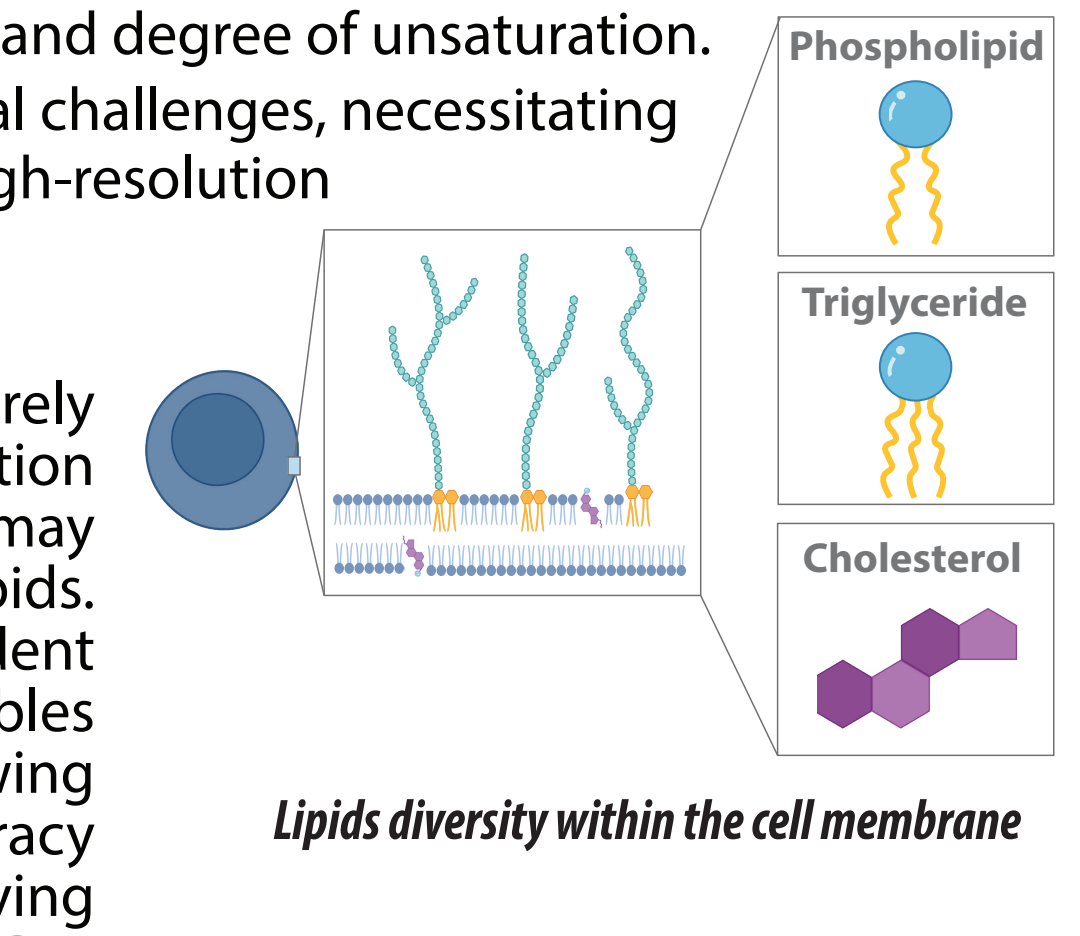
Background

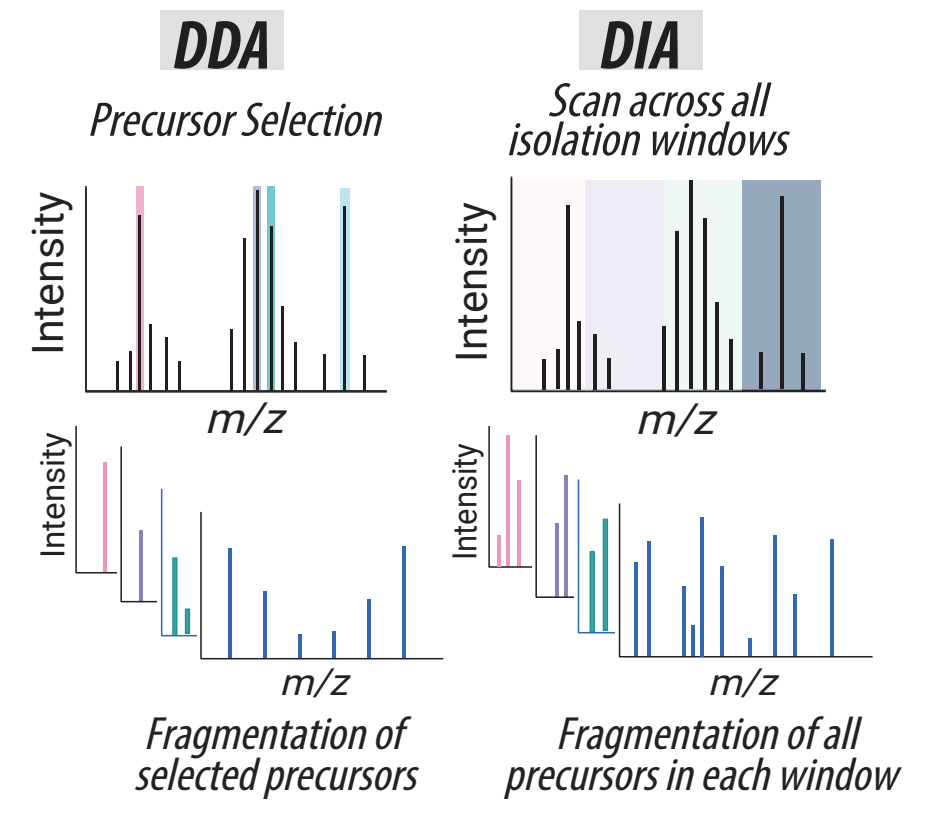
Lipids are remarkably diverse biomolecules, each differing in fatty acid composition, length, and degree of unsaturation. This complexity poses analytical challenges, necessitating advanced technologies for high-resolution and high-sensitivity analysis.



Lipids diversity within the cell membrane

Most lipidomics approaches rely on Data-Dependent Acquisition (DDA) methods, which may overlook low-abundant lipids. Unlike DDA, Data-Independent Acquisition (DIA) enables unbiased ion sampling, allowing for high quantification accuracy for all analytes with varying abundances. However, DIA performance depends on precursor separation efficiency.

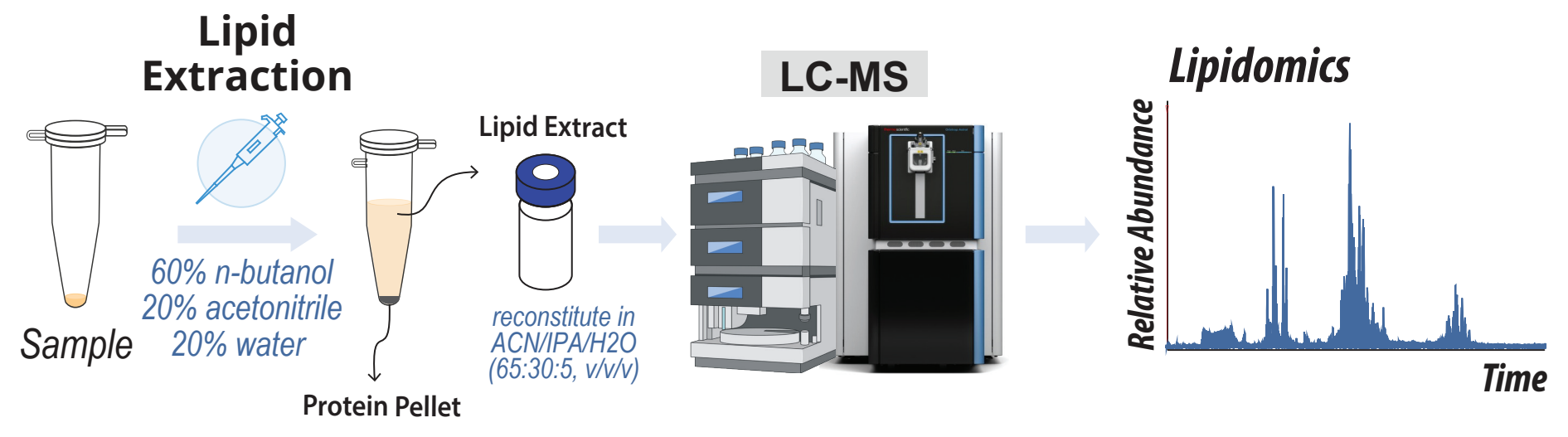
With the introduction of the Astral mass analyzer, offering heightened sensitivity and acquisition rates, we aim to evaluate the performance of DDA and DIA for complex lipid mixtures. By exploring the strengths and limitations of each approach, we seek to develop robust methodologies for a broader set of lipids.


DDA
Precursor Selection
DIA
Scan across all isolation windows
Intensity
m/z
Fragmentation of selected precursors
Fragmentation of all precursors in each window

Key differences between DDA and DIA

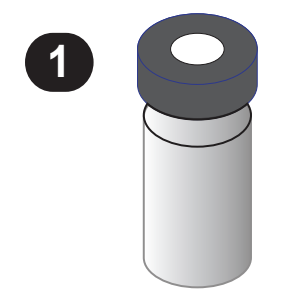
Workflow

Pipeline

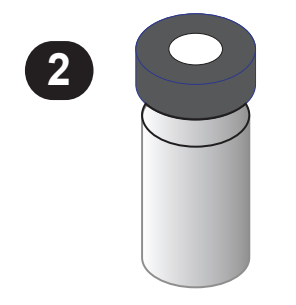


Muehlbauer, L. et. al., Analytical Chemistry (2023)

Samples Used

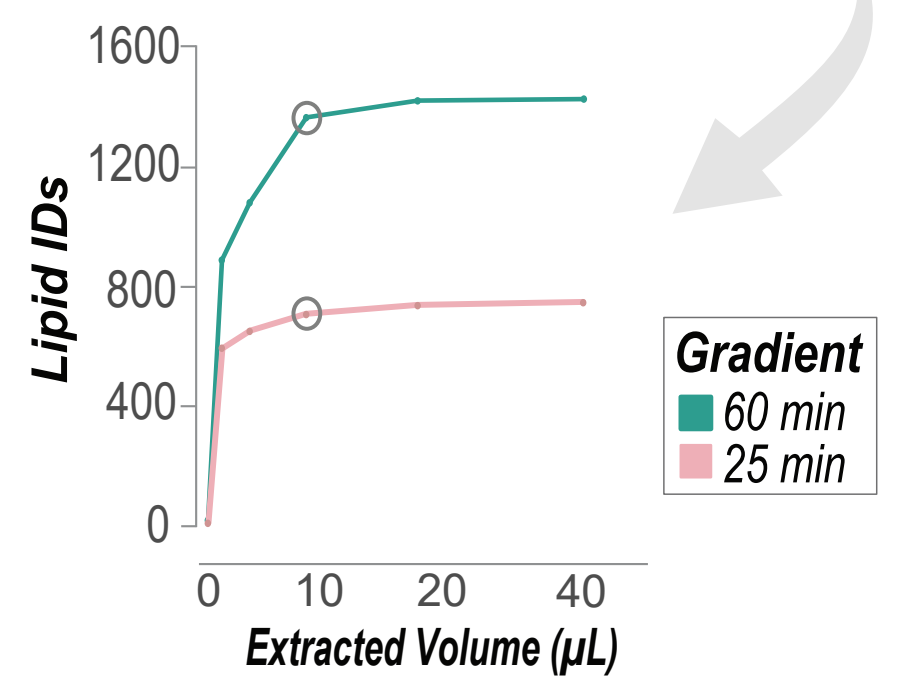


① Ultimate SPLASH (Lipid Standard)



② NIST SRM 1950 (Human Plasma Reference)

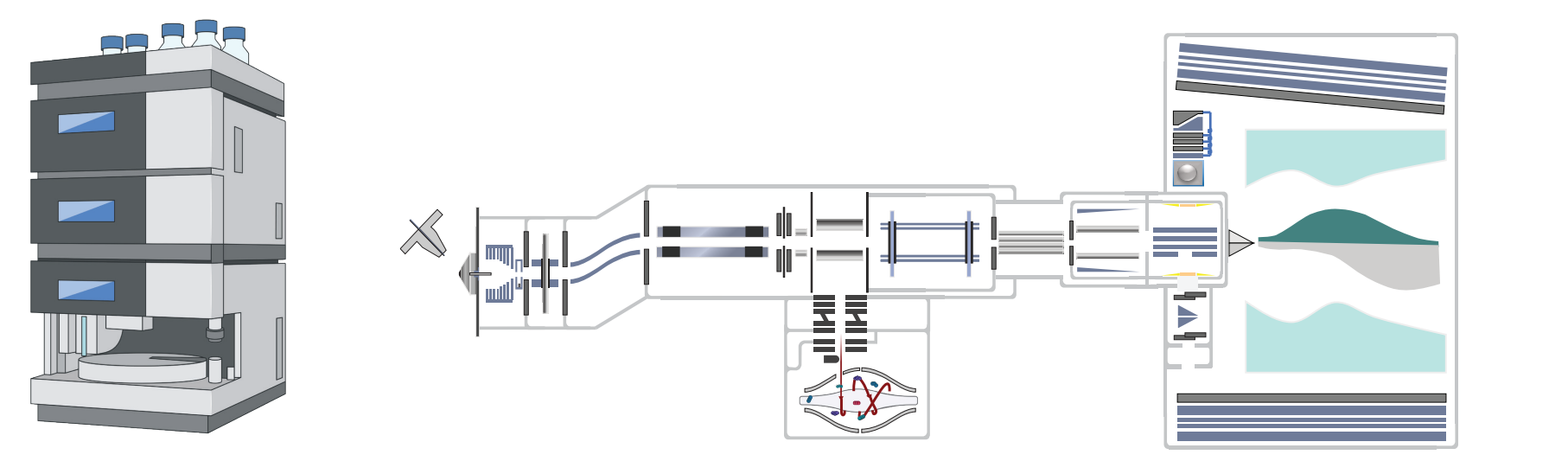
Lipid Class	Count
CE-d7	5
Cer-d7	5
DG-d5	5
LysoPC-d5	3
LysoPE-d5	3
LysoPG-d5	3
LysoPI-d5	3
LysoPS-d5	3
PC-d5	5
PE-d5	5
PG-d5	5
PI-d5	5
PS-d5	5
SM-d9	5
TG-d5	9
Total	69



Different extracted volumes of NIST1950 were evaluated. The number of lipid IDs plateaued with the 10 µL extracted NIST1950. Therefore, we opted to proceed with our experiments utilizing 10 µL of NIST1950 plasma.

Methods and Experiments

Instrumentation



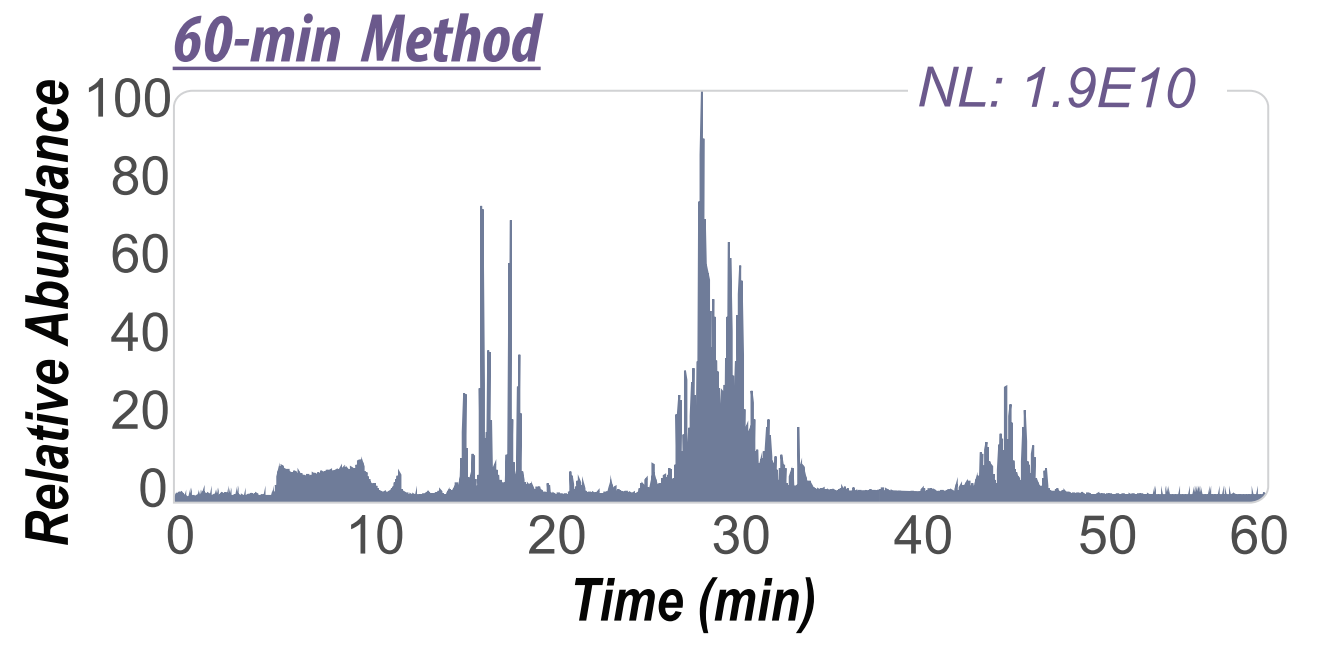
UltiMate 3000 nano HPLC

Hybrid Orbitrap Astral MS Instrument

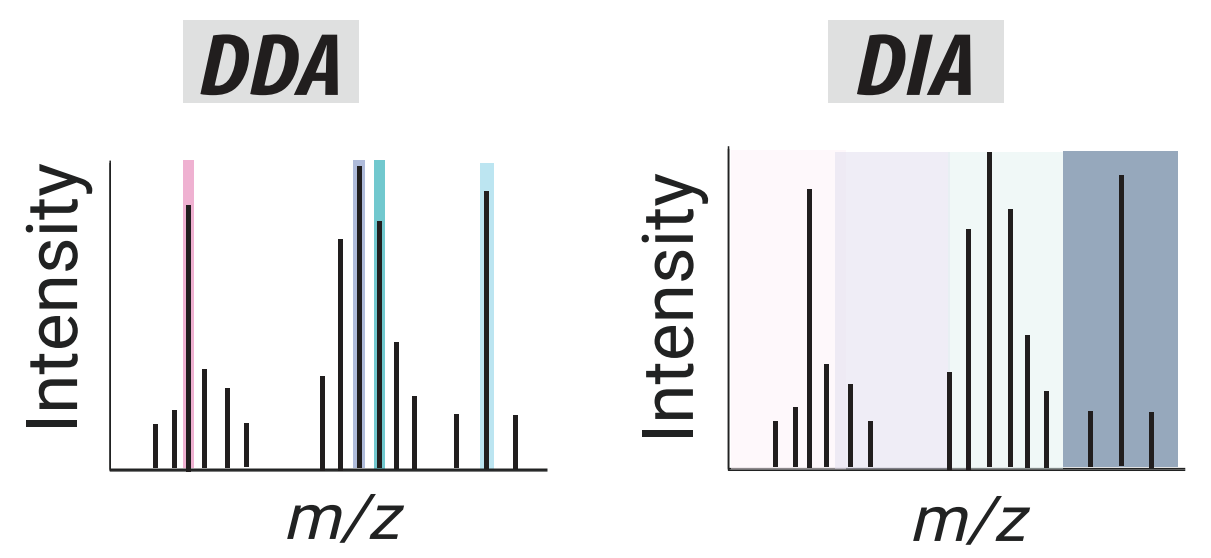
- Lipids were separated using in-house packed C18 column (Waters) with a nano UltiMate 3000 coupled to a Thermo Orbitrap Astral MS (Thermo Scientific). Mobile phases consisted of 0.2% formic acid in water (A), and 0.2% formic acid, 5 mM ammonium formate in (9:1) isoprpropanol /acetonitrile (B).

Variables

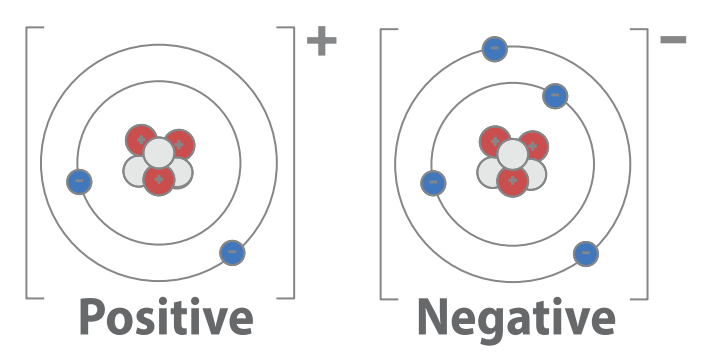
LC-MS Run Time



Acquisition Method



Ionization Mode



MS2 Parameters

Max Injection Time (ms)
Automatic Gain Control (AGC) (%)
Isolation Window for DIA (m/z)

- Separation was accomplished using a stepped gradient of 60-100% mobile phase B across 30 and 11 min for 60-min and 25-min runs, respectively. Data were acquired in positive and negative ion modes with DDA and DIA Astral MS2. MS/MS parameters were optimized, including max injection time (3 ms, 5 ms, 7 ms and 10 ms), (AGC) target% (10, 50, 100, 300, and 500), and isolation windows for DIA (2, 3, 4, 6, 7, and 12 m/z).

Data Analysis

DDA Analysis

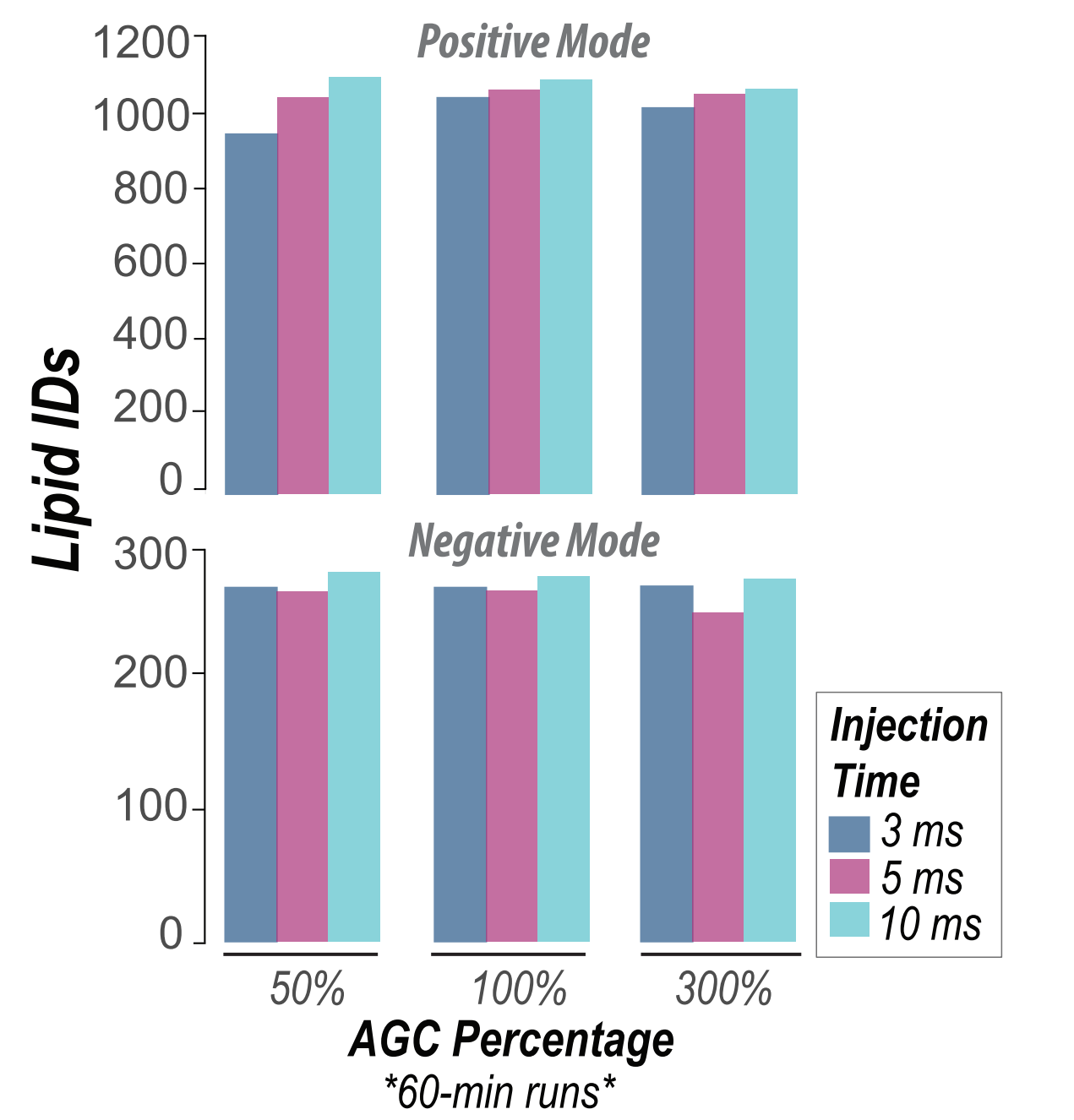
DIA Analysis

- Data were processed using Compound Discoverer 3.3 (Thermo Scientific), LipiDex2 (Anderson, 2024), and MS-DIAL (Tsugawa, 2015).

DDA

AGC% and Injection Time

NIST 1950



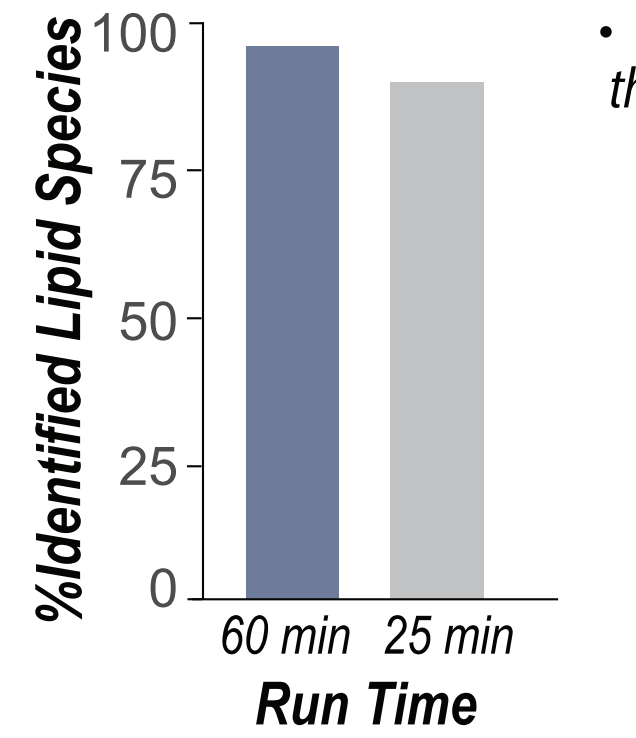
AGC Percentage

60-min runs

- *Remarkably, the 10 ms max injection time consistently outperformed 5 ms and 3 ms, yielding the highest lipid identification on NIST1950. On the other hand, we observed minimal differences with lipid IDs using different AGC% targets.*

Ultimate SPLASH

10 ms Injection Time
100% AGC



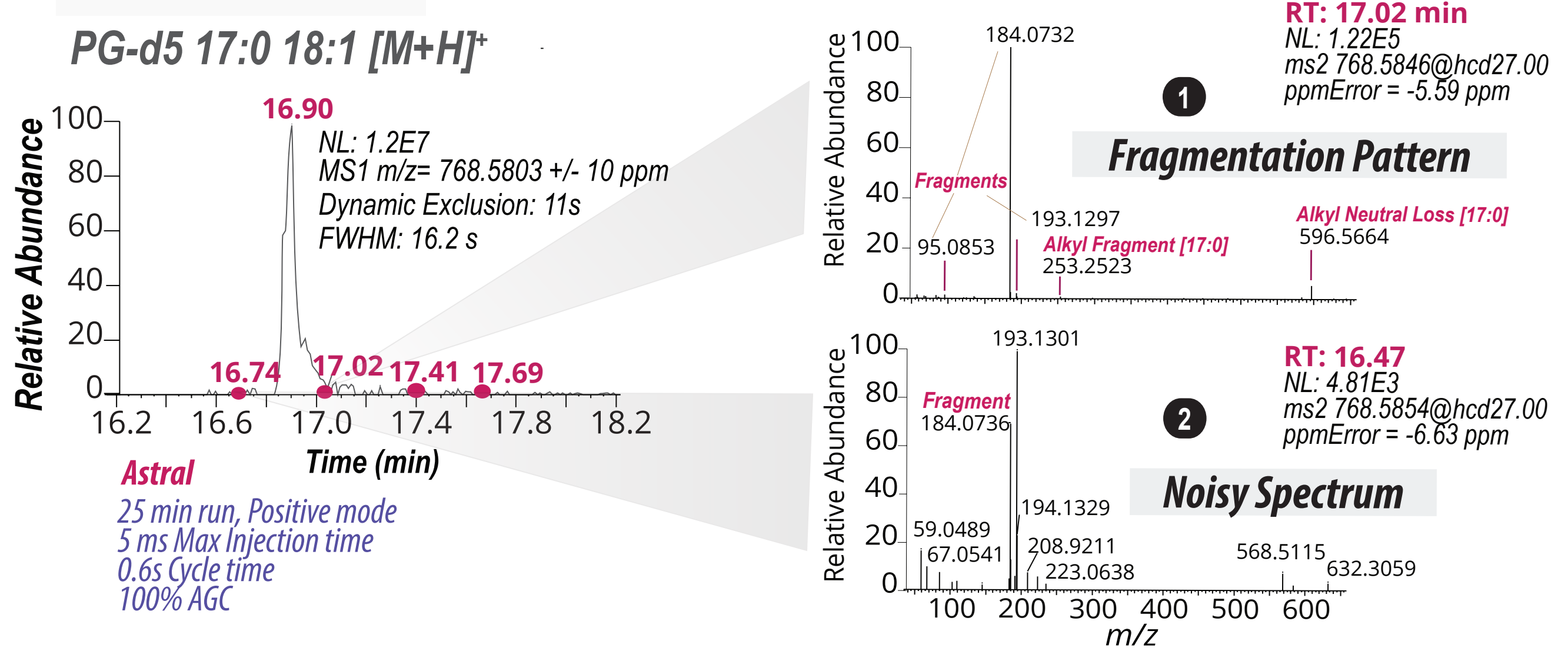
- *93% (64/69) of lipid species in the UltimateSPLASH sample were identified with the 60-minute gradient, compared to 87% (60/69) with the 25-minute gradient.*

Unidentified species were mostly Cholesterol Esters, Phosphoinositides, or Diacylglycerides

Unique DDA Analytical Demands on Astral

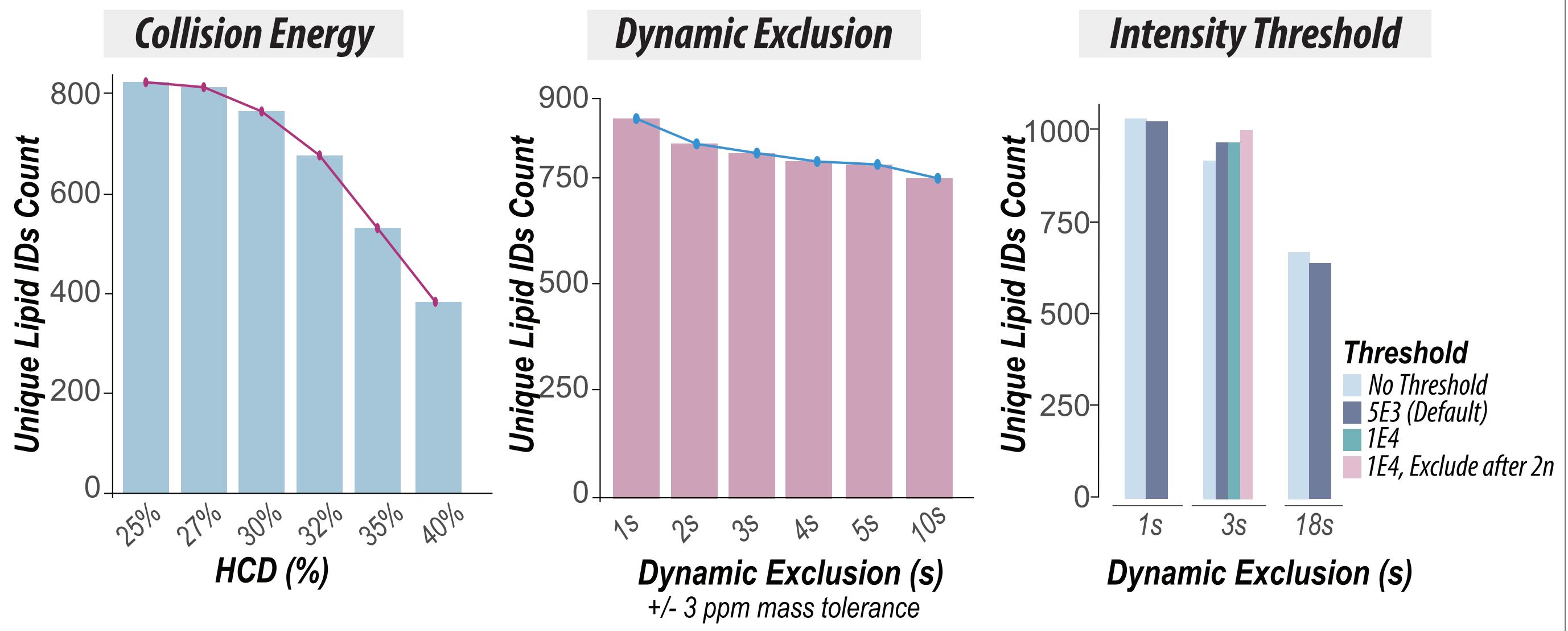
Ultimate SPLASH

PG-d5 17:0 18:1 $[M+H]^+$



- *For unidentified lipid species in the Ultimate SPLASH standard, we found the precursor ions were triggered too early before the main part of the chromatographic peak, due to the high speed scanning of the Astral MS, as shown in this example of Phosphatidylglycerol(PG)-d5 17:0 18:1.*

Method Optimization for Better DDA Sampling

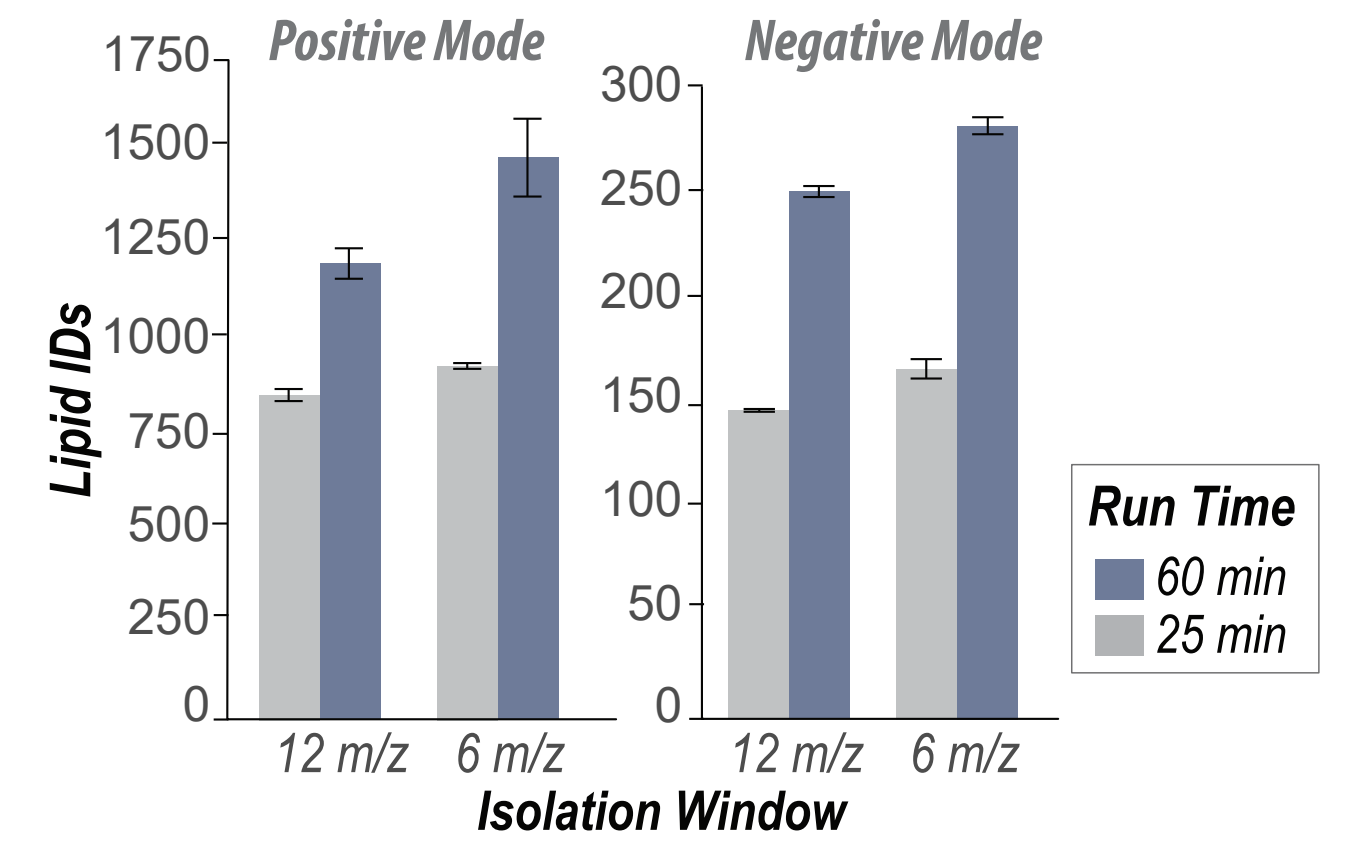


- *Additional method optimization, including different HCD collision energies, dynamic exclusion times, and intensity thresholds, was performed using NIST1950 plasma samples. HCD collision energies of 25% and 27% resulted in the highest number of lipid identifications. The optimal dynamic exclusion time was 1 s. Applying an intensity threshold was generally unhelpful for very high (18 s) or very (1 s) low dynamic exclusion times.*

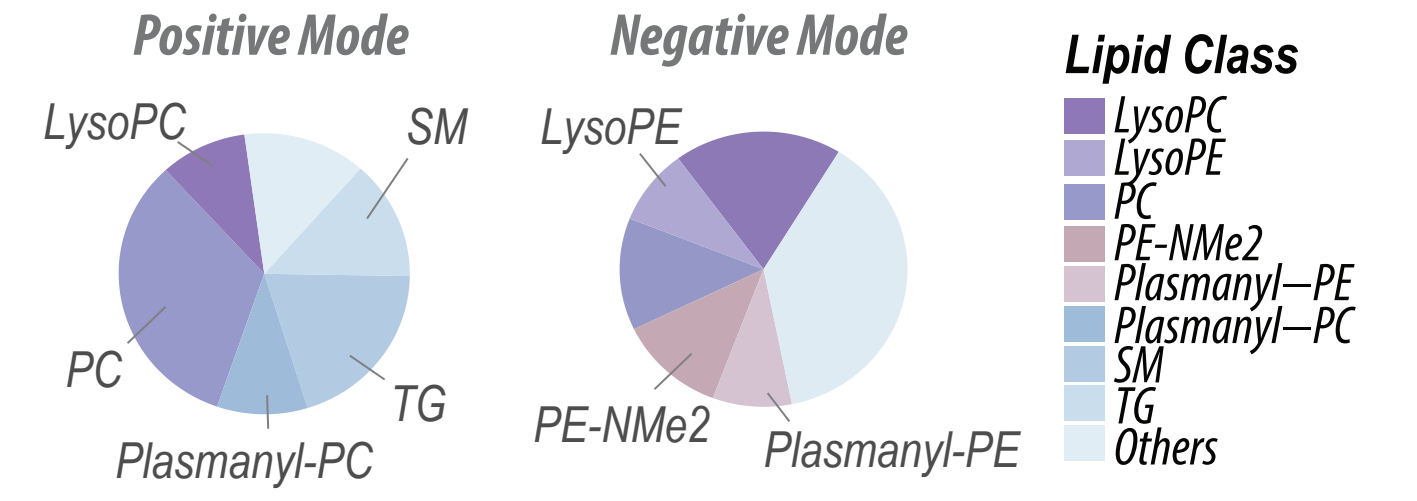
DIA

NIST 1950

Run Time and Isolation Window



- *Exploration of wide (12 m/z) and narrow (6 m/z) isolation window widths favored the 6 m/z window across ionization modes and gradient lengths.*

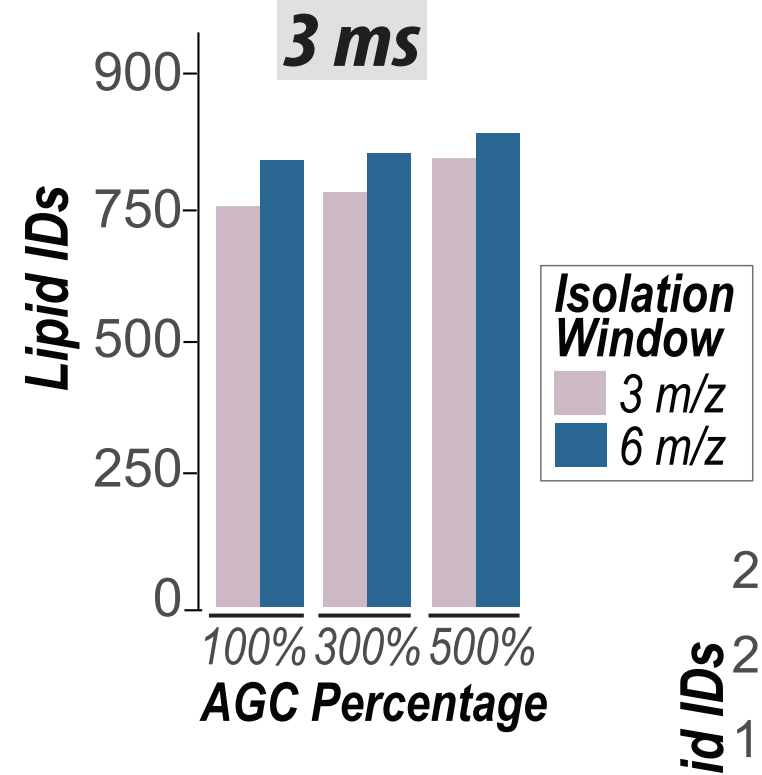


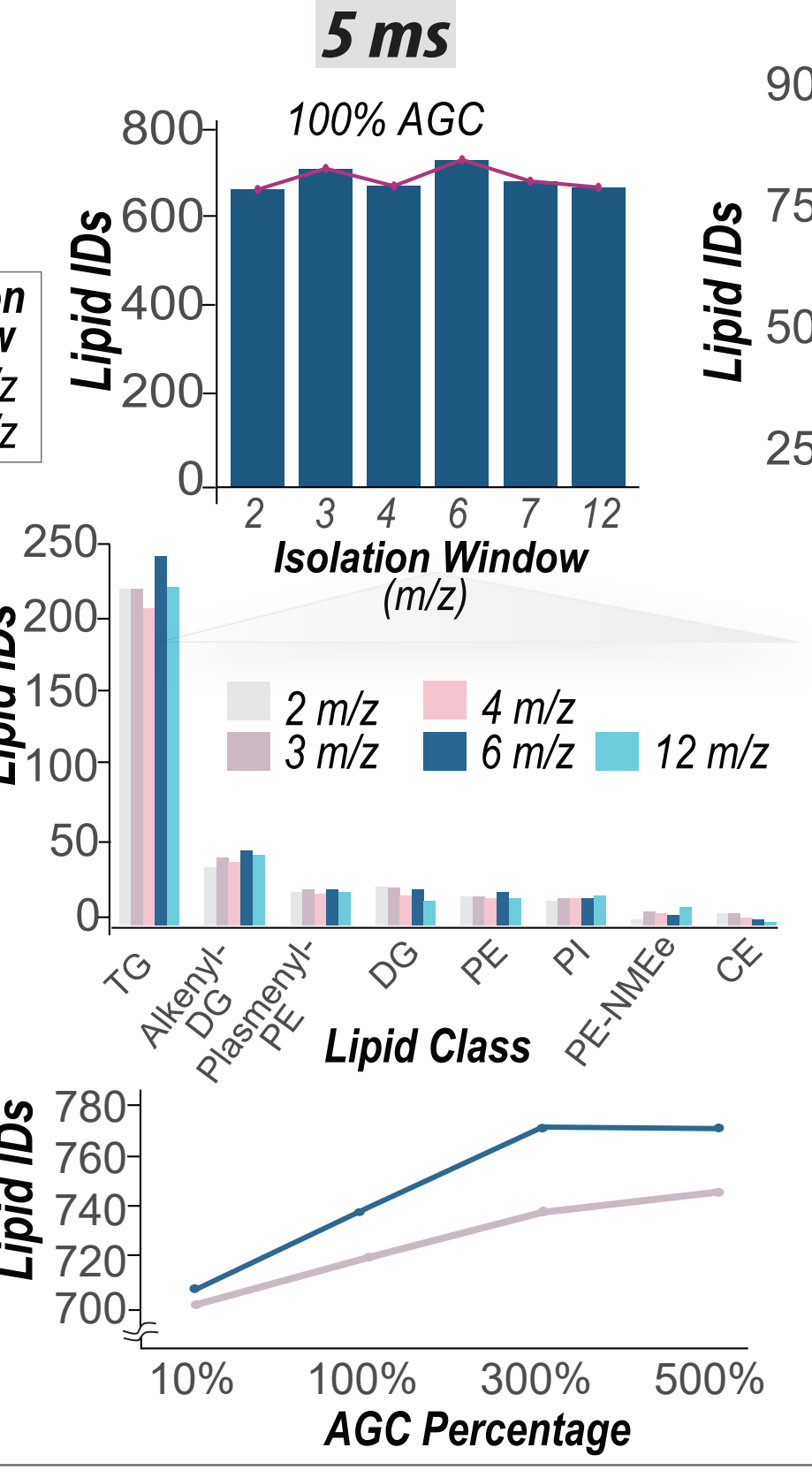
- *Phosphatidylcholines (PCs) and lysophosphatidylcholines (LysoPC) were the most dominant lipid classes in the positive and negative ionization modes, respectively.*

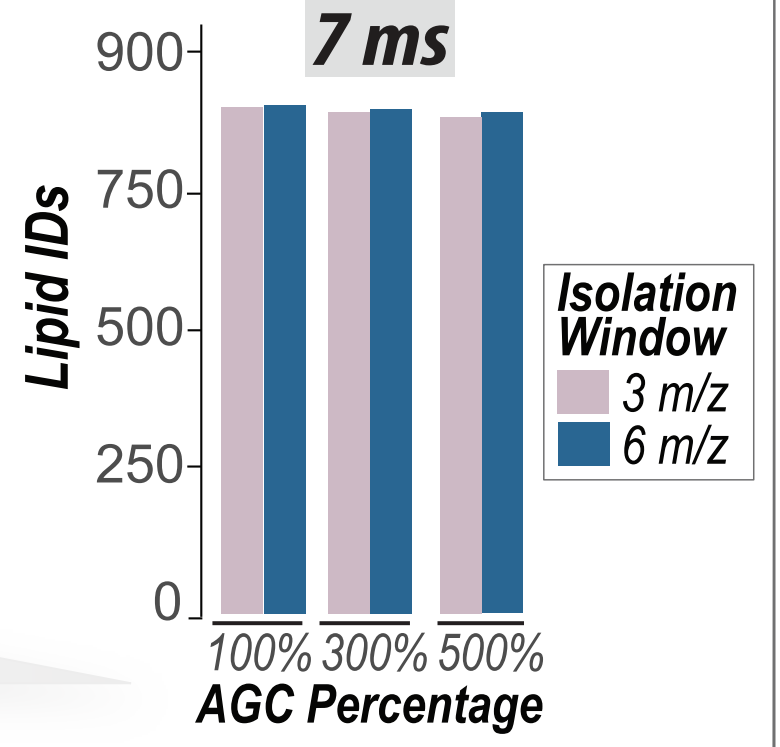
AGC% and Injection Time

- ***Different AGC targets (100%, 300%, and 500%) were evaluated with three Max IT (3 , 5, and 7 ms):***

Note: Some of the dominant lipid classes (i.e., LysoPC, PC, and Sphingomyliens (SMs) were excluded from our MS2 scan range for all the following data.

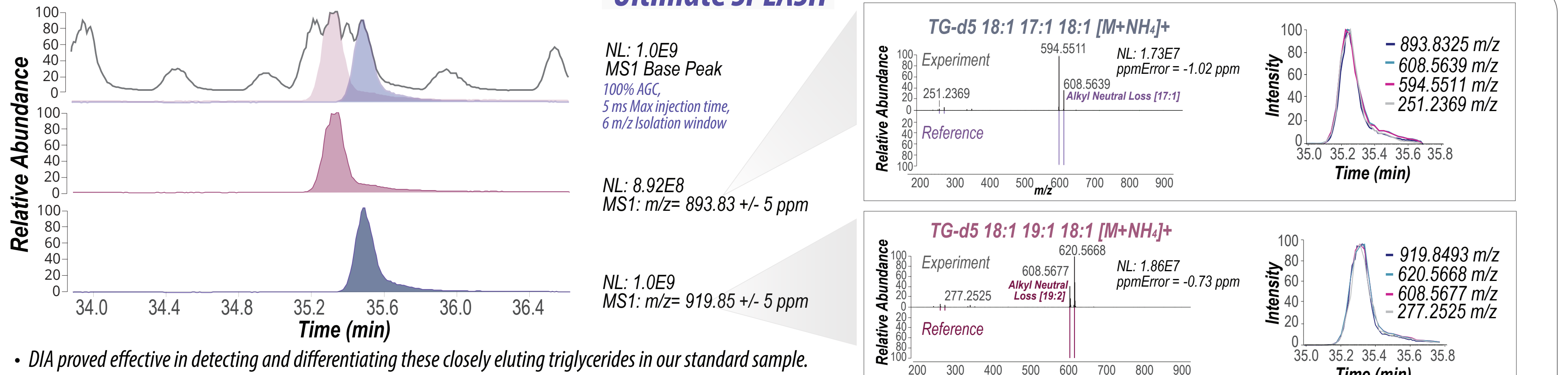






- ***5 ms Injection Time**: Optimizing our parameters with 5 ms max injection time with a fixed 100% AGC and different isolation windows revealed two maxima in lipid identifications at 3 m/z and 6 m/z.*
- *Varying AGC targets (10 to 300%) within these optimal windows demonstrated that lipid identifications increased until saturation at 300% AGC.*
- ***3 ms Injection Time**: Higher AGC targets improved lipid identification, with 500% AGC yielding the best results, followed by 300% and 100%.*
- ***7 ms Injection Time:** No significant differences in lipid identifications were observed among the AGC targets, indicating a plateau effect at this longer injection time.*

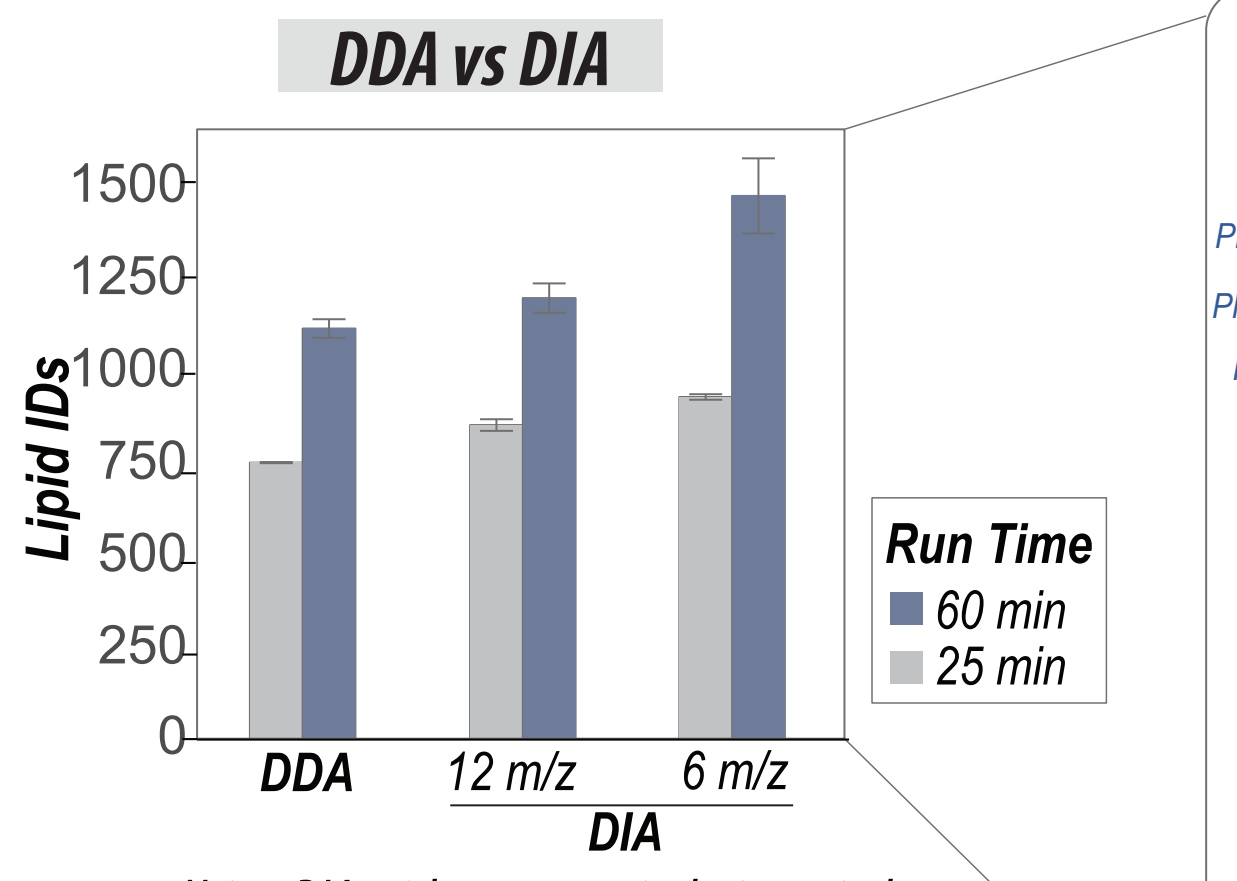
Ultimate SPLASH

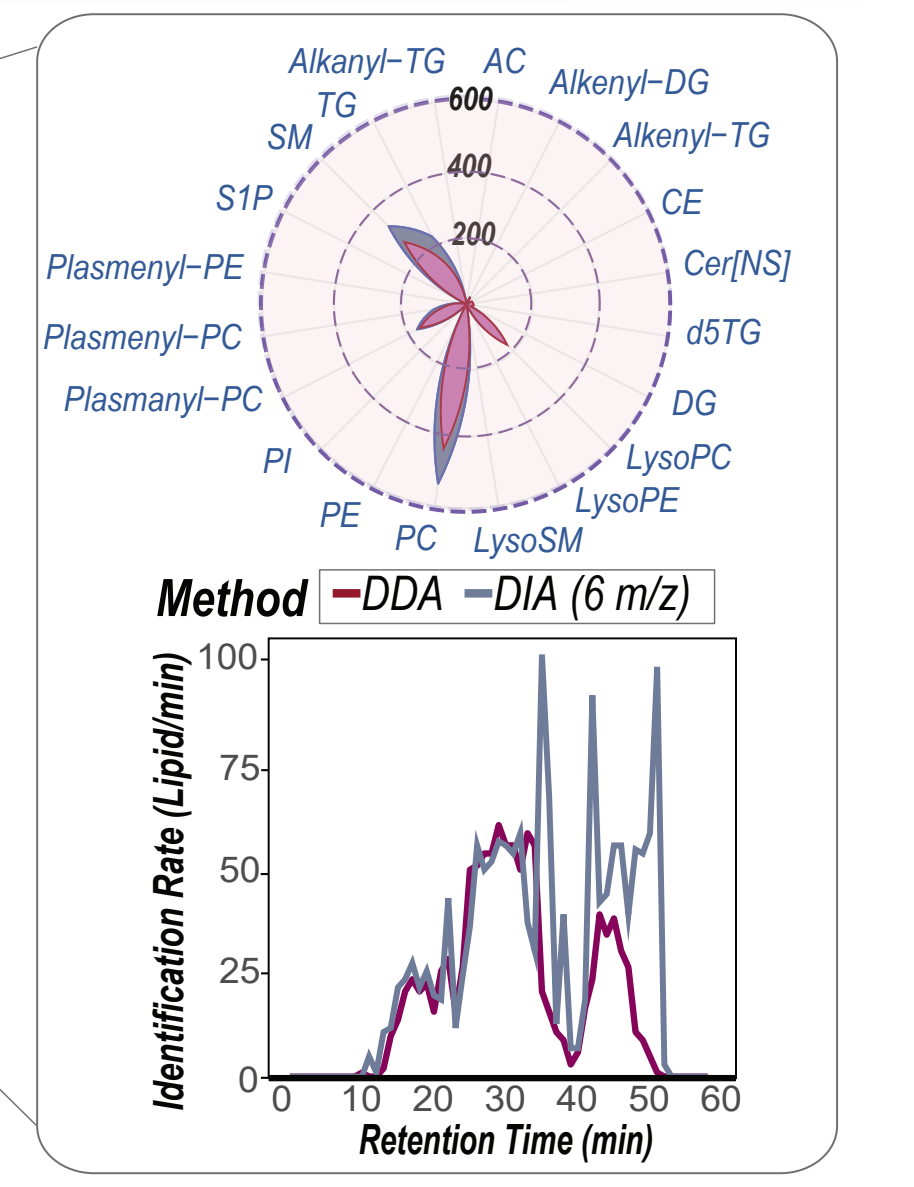


- *DIA proved effective in detecting and differentiating these closely eluting triglycerides in our standard sample. This shows reliable identification and quantification of closely related lipid species using DIA on the Astral MS.*

DDA vs DIA

DDA vs DIA





- *Using DIA with a narrow isolation window (6 m/z) resulted in a 30% and 26% increase in lipid identifications for 60-minute and 25-minute runs, respectively, compared to DDA, with more PCs, SMs and TGs identified.*

Conclusions

- Fine-tuning AGC targets and maximum injection times improved lipid detection while avoiding ion saturation.
- In DDA, short dynamic exclusion times resulted in precursor sampling closer to the peak apex.
- In DIA, narrow isolation windows (3 m/z and 6 m/z) enhanced the identification of most lipid classes.
- DIA identifies more lipids, especially triglycerides and sphingomyelins, than DDA. However, enhanced filtration of peak qualities and signal-to-noise ratios are still imperative to ensure high-quality, reliable lipid identifications, given the current limitations of software processing DIA-Astral data.

Acknowledgments

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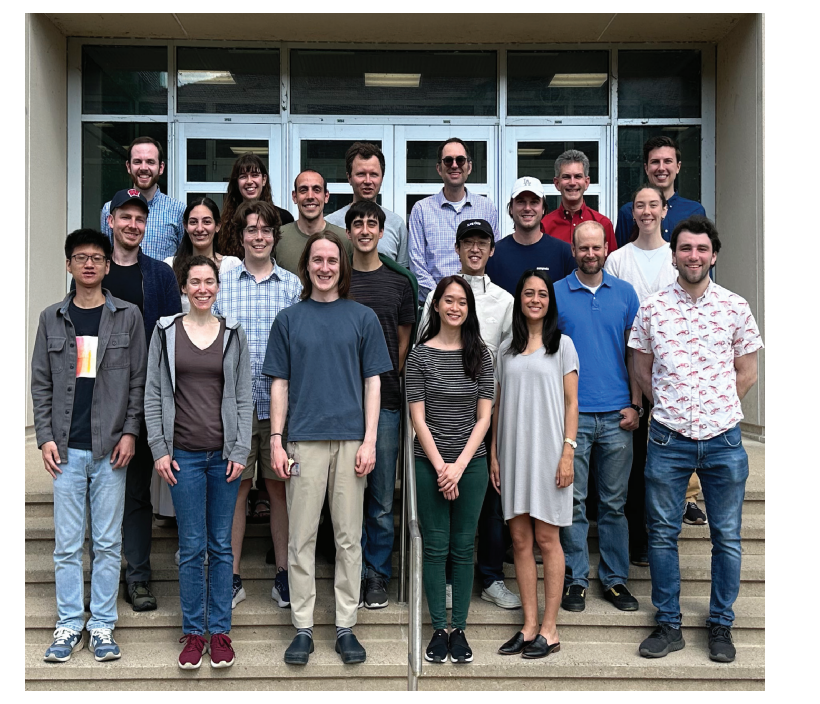
Conflict Statement: JJC is a consultant for Thermo Fisher Scientific, 908 Devices, and Seer.

LinkedIn



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